

Computer Network

Project Report

University Network



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Submitted To:

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Dated: 22th December 2024

Department of Computer Science

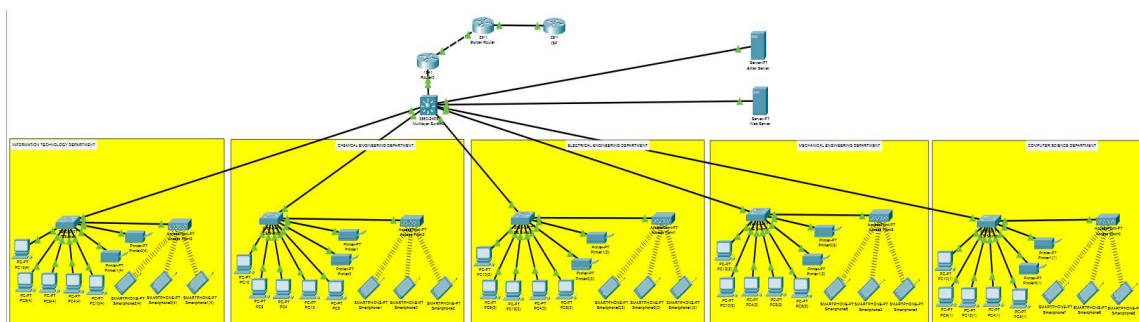
University of Engineering and Technology Lahore, New Campus

Project Specification:

- 1. You are designing a network for:**
University
- 2. No of Separate Units in above mentioned:**
5
- 3. Define the cost or budget of the above mentioned in terms of networking devices:**
No. of Routers: 3
No. of Switches: 6
- 4. Define the number of End Devices allowed in each Unit:**
No. of Personal Computers: 5
Printers: 2
Cellular Devices: 3
Any Other: No
- 5. Define the number of Border Router:**
1
- 6. Define the number of Routers connected to ISP for internet:**
1
- 7. Communication amongst the Units:**
VLAN 10 (CS Department) will be restricted from accessing VLAN 30 (Electrical Engineering Department) and vice versa.
- 8. Define the number of servers and Types:**
2 Servers Used: **Web Server** (HTTP Server) **Email Server** (SMTP/IMAP)
- 9. Un-authorized Access:**
CS & Electrical Department will have access to both servers (Web and Email).
Mechanical Department will not be able to access any Server.
Chemical Department will only be able to access the Web Server.
IT Department will only be able to access the Email Server.
- 10. Routing Protocols used:**
Between internal routers: Static Routing
With ISP router: Static Routing
- 11. Private IP used (select one prefix):**
We are using 192.168/16 private IP.
- 12. No. of Public Addresses allocated by ISP to above mentioned:**
The ISP has allocated 10 public IPs (105.12.32.15–105.12.32.25), with 105.12.32.24 reserved for the Email Server, 105.12.32.23 for the Web Server, and all internal devices using Static NAT for Internet access via the Border Router.

Report:

1. Diagram (Topology)



2. Sub netting

a. VLISM

Network	Subnet	Prefix	Subnet Mask	Hosts	IP Range	Broadcast
CS	192.168.1.0	/28	255.255.255.240	14	192.168.1.1–192.168.1.14	192.168.1.15
ME	192.168.2.0	/28	255.255.255.240	14	192.168.2.1–192.168.2.14	192.168.2.15
EE	192.168.3.0	/28	255.255.255.240	14	192.168.3.1–192.168.3.14	192.168.3.15
CE	192.168.4.0	/28	255.255.255.240	14	192.168.4.1–192.168.4.14	192.168.4.15
IT	192.168.5.0	/28	255.255.255.240	14	192.168.5.1–192.168.5.14	192.168.5.15
Servers	192.168.6.0	/29	255.255.255.248	6	192.168.6.1–192.168.6.6	192.168.6.7
Link	192.168.7.0	/30	255.255.255.252	2	192.168.7.1–192.168.7.2	192.168.7.3

b. VLAN

VLAN ID	VLAN Name	Network	Assigned Devices
10	CS_VLAN	192.168.1.0/28	PCs, Smartphones, Printers in CS Department
20	ME_VLAN	192.168.2.0/28	PCs, Smartphones, Printers in ME Department
30	EE_VLAN	192.168.3.0/28	PCs, Smartphones, Printers in EE Department
40	CE_VLAN	192.168.4.0/28	PCs, Smartphones, Printers in CE Department
50	IT_VLAN	192.168.5.0/28	PCs, Smartphones, Printers in IT Department
60	VLAN_Server	192.168.6.0/29	Web and Email Server

3. Configurations

a. Switch

One multi-layer switch used for Inter-VLAN Routing between the VLANs.

syeds\OneDrive\...
Tools Exten
x: 898, y: 252

Physical Config CLI Attributes

IOS Command Line Interface

```
!
version 12.2(37)SE1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
ip dhcp pool DeptA_DHCP
network 192.168.1.0 255.255.255.240
default-router 192.168.1.1
dns-server 8.8.8.8
ip dhcp pool DeptB_DHCP
network 192.168.2.0 255.255.255.240
default-router 192.168.2.1
dns-server 8.8.8.8
!
!
ip routing
!
!
!
!
!
```

syeds\OneDrive\...
Tools Exten
x: 1089, y: 259

Physical Config CLI Attributes

IOS Command Line Interface

```
!
interface FastEthernet0/1
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface FastEthernet0/2
switchport mode dynamic desirable
!
interface FastEthernet0/3
switchport mode dynamic desirable
!
interface FastEthernet0/4
switchport mode dynamic desirable
!
interface FastEthernet0/5
switchport mode dynamic desirable
!
interface FastEthernet0/6
switchport mode dynamic desirable
!
interface FastEthernet0/7
switchport access vlan 60
switchport mode access
switchport nonegotiate
!
interface FastEthernet0/8
switchport access vlan 60
switchport mode access
switchport nonegotiate
!
interface FastEthernet0/9
switchport access vlan 60
switchport mode access
switchport nonegotiate
!
!
```

Ctrl+F6 to exit CLI focus

Copy Paste

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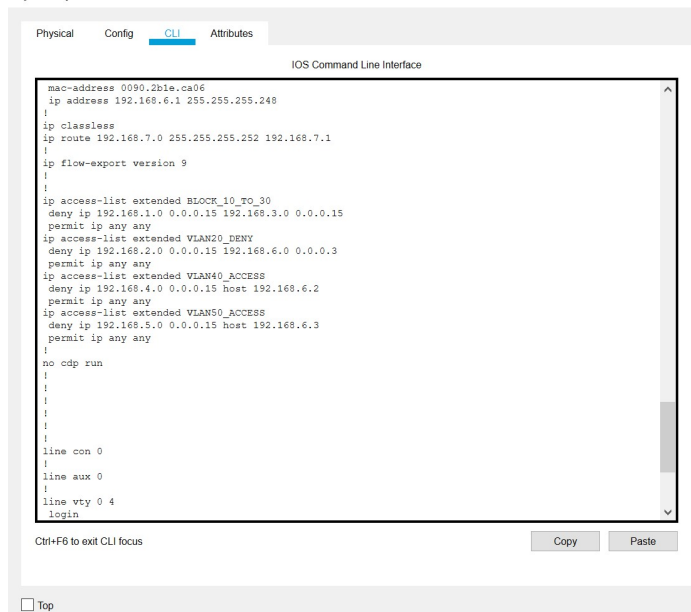
Physical Config CLI Attributes

IOS Command Line Interface

```
!
interface Vlan1
no ip address
shutdown
!
interface Vlan10
mac-address 0090.2b1e.ca01
ip address 192.168.1.1 255.255.255.240
ip access-group BLOCK_10_TO_30 in
!
interface Vlan20
mac-address 0090.2b1e.ca02
ip address 192.168.2.1 255.255.255.240
ip access-group VLAN20_DENY in
!
interface Vlan30
mac-address 0090.2b1e.ca03
ip address 192.168.3.1 255.255.255.240
ip access-group BLOCK_10_TO_30 in
!
interface Vlan40
mac-address 0090.2b1e.ca04
ip address 192.168.4.1 255.255.255.240
ip access-group VLAN40_ACCESS in
!
interface Vlan50
mac-address 0090.2b1e.ca05
ip address 192.168.5.1 255.255.255.240
ip access-group VLAN50_ACCESS in
!
interface Vlan60
mac-address 0090.2b1e.ca06
ip address 192.168.6.1 255.255.255.240
!
ip classless
!
```

Ctrl+F6 to exit CLI focus

Copy Paste



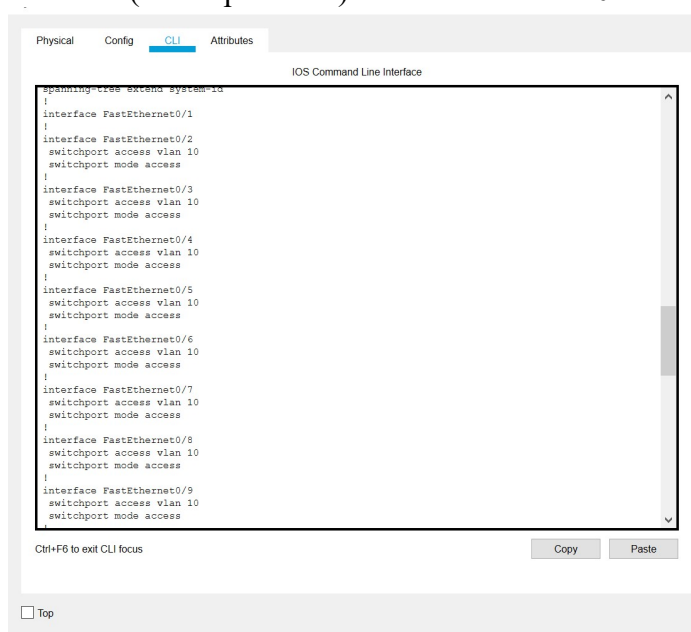
```
Physical Config CLI Attributes
IOS Command Line Interface

mac-address 0090.2b1e.ca06
ip address 192.168.6.1 255.255.255.248
!
ip classless
ip route 192.168.7.0 255.255.255.252 192.168.7.1
!
ip flow-export version 9
!
!
ip access-list extended BLOCK_10_TO_30
deny ip 192.168.1.0 0.0.0.15 192.168.3.0 0.0.0.15
permit ip any any
ip access-list extended VLAN20_DENY
deny ip 192.168.2.0 0.0.0.15 192.168.6.0 0.0.0.3
permit ip any any
ip access-list extended VLAN40_ACCESS
deny ip 192.168.4.0 0.0.0.15 host 192.168.6.2
permit ip any any
ip access-list extended VLAN50_ACCESS
deny ip 192.168.5.0 0.0.0.15 host 192.168.6.3
permit ip any any
!
no cdp run
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login

Ctrl+F6 to exit CLI focus Copy Paste
```

☐ Top

Switch 1 (CS Department) will have VLAN 10.



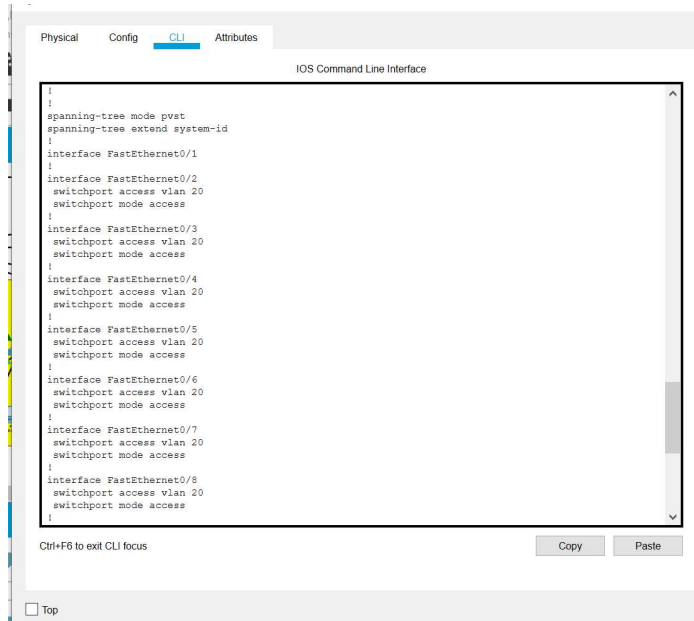
```
Physical Config CLI Attributes
IOS Command Line Interface

!
!
interface FastEthernet0/1
!
interface FastEthernet0/2
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/3
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/4
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/5
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/6
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/7
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/8
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 10
switchport mode access
!

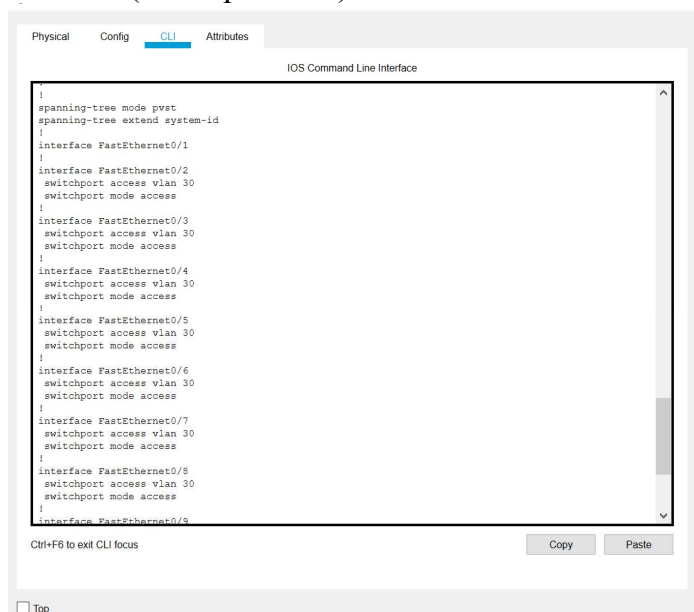
Ctrl+F6 to exit CLI focus Copy Paste
```

☐ Top

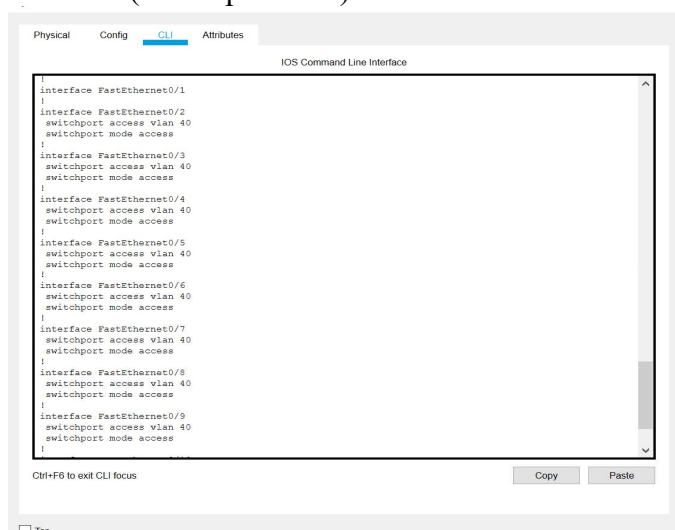
Switch 2 (ME Department) will have VLAN 20.



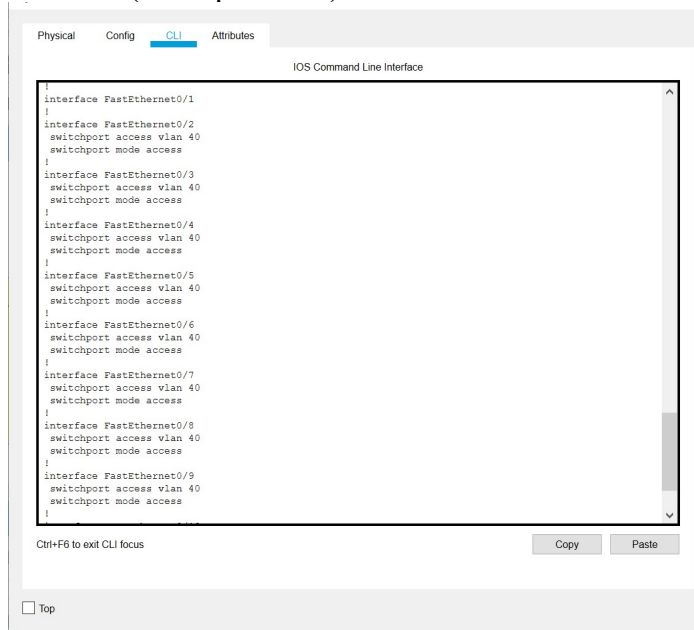
Switch 3 (EE Department) will have VLAN 30.



Switch 4 (CE Department) will have VLAN 40.

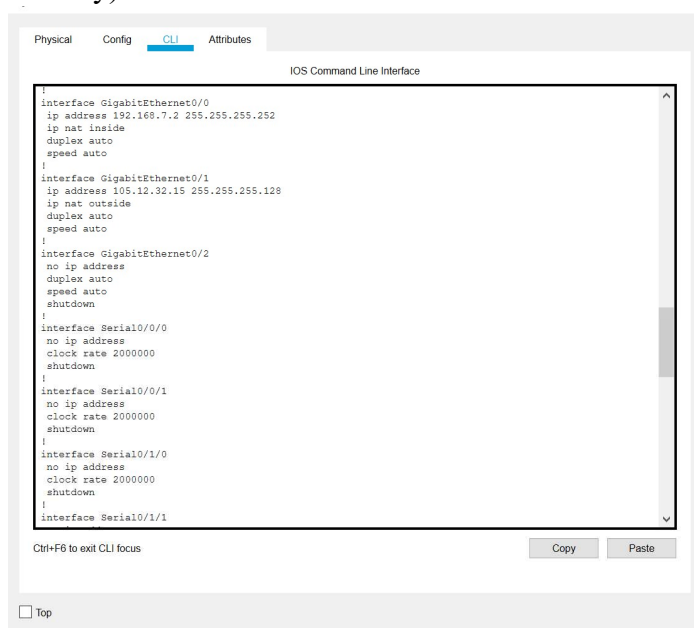


Switch 5 (IT Department) will have VLAN 50.



b. Router

Border Router (configured with static routing, NAT, and access lists for security).



```

!
interface Vlan1
  no ip address
  shutdown
!
ip nat pool ISP_POOL 105.12.32.17 105.12.32.22 netmask 255.255.255.128
ip nat inside source list 1 pool ISP_POOL overload
ip nat inside source static 192.168.6.3 105.12.32.23
ip nat inside source static 192.168.6.2 105.12.32.24
ip classless
ip route 192.168.1.0 255.255.255.240 192.168.7.1
ip route 192.168.2.0 255.255.255.240 192.168.7.1
ip route 192.168.3.0 255.255.255.240 192.168.7.1
ip route 192.168.4.0 255.255.255.240 192.168.7.1
ip route 192.168.5.0 255.255.255.240 192.168.7.1
ip route 192.168.6.0 255.255.255.248 192.168.7.1
ip route 0.0.0.0 0.0.0.0 105.12.32.16
!
ip flow-export version 9
!
!
access-list 1 permit 192.168.0.0 0.0.255.255
!
no cdp run
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
!

```

Internal Routers (connected to the multi-layer switch to handle inter-VLAN communication).

```

no ip address
duplex auto
speed auto
!
interface FastEthernet0/0.10
  encapsulation dot1Q 10
  ip address 192.168.1.1 255.255.255.240
!
interface FastEthernet0/0.20
  encapsulation dot1Q 20
  ip address 192.168.2.1 255.255.255.240
!
interface FastEthernet0/0.30
  encapsulation dot1Q 30
  ip address 192.168.3.1 255.255.255.240
!
interface FastEthernet0/0.40
  encapsulation dot1Q 40
  ip address 192.168.4.1 255.255.255.240
!
interface FastEthernet0/0.50
  encapsulation dot1Q 50
  ip address 192.168.5.1 255.255.255.240
!
interface FastEthernet0/0.60
  encapsulation dot1Q 60
  ip address 192.168.6.1 255.255.255.248
!
interface FastEthernet0/1
  ip address 192.168.7.1 255.255.255.252
  duplex auto
  speed auto
!
interface Vlan1
  no ip address
  shutdown
!

```

4. Simulations

• Test 1: Communication b/w the Units

Units	Expected Result	Actual Result	Test Result
CS to ME	Can communicate	As expected	Pass
CS to EE	Cannot communicate	As expected	Pass
CS to CE	Can communicate	As expected	Pass
CS to IT	Can communicate	As expected	Pass
ME to CS	Can communicate	As expected	Pass

ME to EE	Can communicate	As expected	Pass
ME to CE	Can communicate	As expected	Pass
ME to IT	Can communicate	As expected	Pass
EE to CS	Cannot communicate	As expected	Pass
EE to ME	Can communicate	As expected	Pass
EE to CE	Can communicate	As expected	Pass
EE to IT	Can communicate	As expected	Pass
CE to CS	Can communicate	As expected	Pass
CE to ME	Cannot communicate	As expected	Pass
CE to EE	Can communicate	As expected	Pass
CE to IT	Can communicate	As expected	Pass
IT to CS	Can communicate	As expected	Pass
IT to ME	Can communicate	As expected	Pass
IT to EE	Can communicate	As expected	Pass
IT to CE	Can communicate	As expected	Pass

CS Department communication with Mechanical Department

```
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=1ms TTL=127
Reply from 192.168.2.3: bytes=32 time=12ms TTL=127
Reply from 192.168.2.3: bytes=32 time=3ms TTL=127
Reply from 192.168.2.3: bytes=32 time=3ms TTL=127

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 4ms
```

Mechanical Department communication with Electrical Department

```
Command Prompt

Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.4

Pinging 192.168.3.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.4: bytes=32 time<1ms TTL=127
Reply from 192.168.3.4: bytes=32 time=11ms TTL=127
Reply from 192.168.3.4: bytes=32 time=14ms TTL=127

Ping statistics for 192.168.3.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 8ms

C:\>
```

Department CS cannot communicate with Electrical Department

```

C:\>ping 192.168.3.4

Pinging 192.168.3.4 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

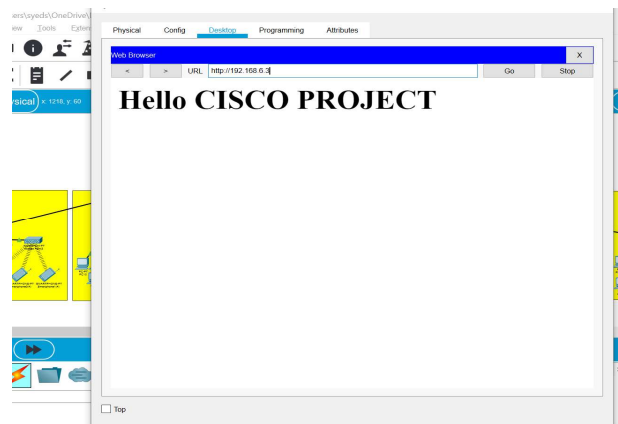
Ping statistics for 192.168.3.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

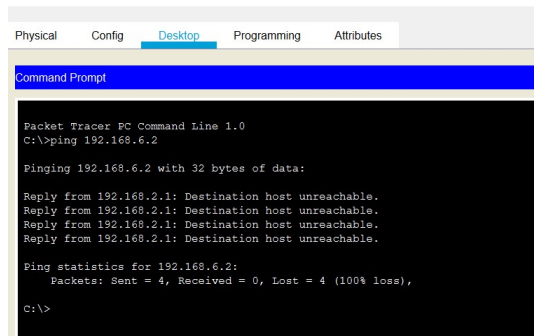
- **Test 2: Server Accesses**

Units to Server	Expected Result	Actual Result	Test Result
CS to Web Server	Can communicate	As expected	Pass
CS to Email Server	Can communicate	As expected	Pass
ME to Web Server	Cannot communicate	As expected	Pass
ME to Email Server	Cannot communicate	As expected	Pass
EE to Web Server	Can communicate	As expected	Pass
EE to Email Server	Can communicate	As expected	Pass
CE to Web Server	Can communicate	As expected	Pass
CE to Email Server	Cannot communicate	As expected	Pass
IT to Web Server	Cannot communicate	As expected	Pass
IT to Email Server	Can communicate	As expected	Pass

CS Department Access Web Server



Chemical Department can't access Email Server



Physical Config **Desktop** Programming Attributes

Command Prompt

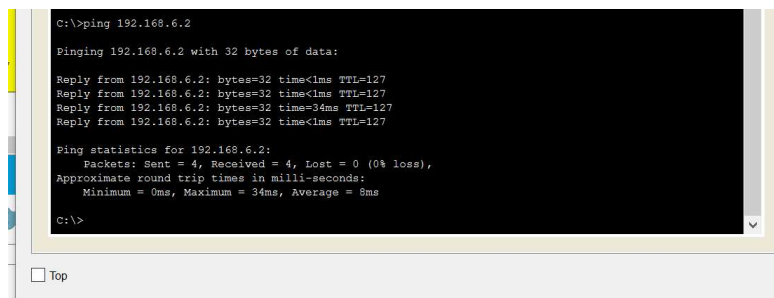
```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.6.2

Pinging 192.168.6.2 with 32 bytes of data:

Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.

Ping statistics for 192.168.6.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
```

IT Department can access Email Server



```
C:\>ping 192.168.6.2

Pinging 192.168.6.2 with 32 bytes of data:

Reply from 192.168.6.2: bytes=32 time<1ms TTL=127
Reply from 192.168.6.2: bytes=32 time<1ms TTL=127
Reply from 192.168.6.2: bytes=32 time=34ms TTL=127
Reply from 192.168.6.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.6.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 34ms, Average = 8ms
C:\>
```

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- **Test for ISP:**
All the switches can connect to ISP through NAT protocol.